

The Talk: Speculative Conversation with Everyday Objects

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ABSTRACT

Communication between humans and everyday objects is often implicit. In this paper, we create speculative scenarios where users could have a conversation, as explicit communication, with everyday objects that usually may not have conversational artificial intelligence (AI) installed, such as a book. We present a design fiction narrative and a conversational system about conversations with everyday objects. Our user test showed a positive acceptance of conversation with everyday objects, and users tried to build a human-like relationship with them.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in interaction design**.

KEYWORDS

conversational AI, design fiction narrative, speculative design, everyday objects

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1 INTRODUCTION

Communication between humans and everyday objects has been investigated in many ways to explore expressive and behavioral potentials. For example, a user can control a blender by imitating a blender's voice [Dobson 2004]. A lamp captures a user's movement and communicates with a user by its movement [Ben-Dror et al. 2012]. These projects emphasize empathic and implicit communication. Verbal communication is often regarded as a more explicit means of communication [Lausic 2005], compared to implicit communication methods such as voice volume. Artificial intelligence (AI) assistants, such as Amazon Alexa, enable users to have verbal

communication with objects but they have limited context understanding. Users must specify actions, for example, "Alexa, turn on the living room light.", as Alexa cannot comprehend objects and surroundings which users want to control. Incorporating the user's context could streamline their interaction by simply saying "Turn on." The context can omit some sentences to explain details. Moreover, accessibility can be enhanced by directly talking to everyday objects that are located where users interact, while AI assistant devices are available only in specific locations. Communication with AI assistants is limited to simple request and response interactions [Reeves et al. 2018], but AI has progressed to have better conversational qualities. The movie *Her* [Jonze 2013] suggests that there is a possibility to create an emotional connection with conversational AI and it could become your partner or friend. Furthermore, it highlights that this emotional connection helps reduce a sense of social isolation. Conversing directly and explicitly with everyday objects could omit an explanation of contexts, enhance an accessibility and create emotional connections with everyday objects.

We created speculative scenarios where users can engage in conversations with everyday objects, specifically a glass and a travel book. We chose a glass as an example due to its close proximity and frequent interaction with users, making it a personal object even if you don't own it. Additionally, we selected a travel book due to its clear indication of the common topic by its cover. We conducted a user test utilizing a design fiction narrative and a conversational system to understand whether users accept engaging in conversation with everyday objects and how they perceive these interactions.

2 RELATED WORKS

2.1 Conversation with Everyday Objects

Speculative conversations between humans and everyday objects has been previously studied to understand how objects perceive during conversations [Reddy 2021]. Participants role-played as objects, influencing conversation content with their knowledge. Our project investigates how humans perceive and react through the functional prototype which enables more informed and knowledgeable conversations.

2.2 Integration of Conversational AI

Since ChatGPT was released in November 2022, people have started integrating it to have human-like conversations with their favorite characters [inews 2023; twitter 2023]. In our project, we focus on a

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conversation between humans and everyday objects, not characters or AI assistants.

3 SYSTEM DESIGN AND IMPLEMENTATION

To understand how users feel and react to conversations with everyday objects from a human perspective, we created a design fiction narrative and a conversational system. The narrative fostered empathy for the main character's emotion to objects, while the system enabled direct conversations, allowing users to discuss various topics with objects.

3.1 Design Fiction Narrative

This is the summary of the design fiction narrative.

Aria, living in 2040, uses an AR contact lens which enables her to have conversations with everyday objects. She visits a bar and converses with a glass. It brings back memories of her father who passed away, evoking a sense of nostalgia and loss.

3.2 Conversational System

A conversational system called The Talk was created in the form of a travel book, which we selected due to its clear indication of the topic by its cover. We assumed that users wouldn't run out of things to talk about during conversation. We place the book on the desk, where a projector shows a visual effect to help users to understand which object is talking to them and the conversation status ('Idle' and 'Talking in progress').

Whisper and ChatGPT [OpenAI 2023] are implemented to recognize a user's voice and generate an answer from a user's sentence. The generated sentence is converted to voice by Amazon Polly [Amazon 2023].

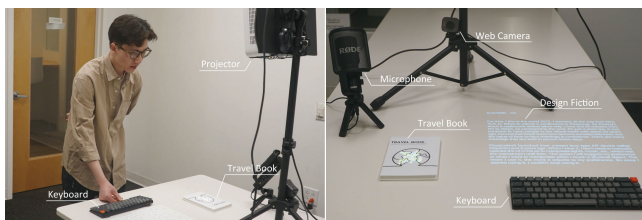


Figure 1: The setup of the conversational system, The Talk

4 USER TEST AND RESULTS

4.1 User Test Design

User tests were conducted with 5 people (4 females and 1 male; $n = 5$, mean age = 25.2). Each user was asked to read the design fiction narrative and talk with the travel book about any topic for 8 minutes. Semi-structured interviews were then conducted to understand users' acceptance of conversing with everyday objects and their perceptions of these interactions. The interviews focused on their feelings during conversation, the reason behind their specific sentences to the book, their impressions of the design fiction narrative and other everyday objects they want to converse with.

4.2 Major Insights

All users enjoyed conversing with the travel book and continued their conversations. Person A said "Reading is not my style so talking is a better option for me." Person B said "Speech is easy and casual but depends on the situation." While conversation is not the only option, users felt talking to a book was acceptable.

Some users attempted to build a human-like relationship with the book. Person C said "Thank you, Travel Book." after the book told her travel ideas. She thought those words, such as thank you and calling its name, would make the conversation with a book smoother. Person D said "He would be my friend." Users instinctively use words that are normally used towards humans.

All users stayed focused on a travel topic during the conversation with the book. Person C said "I assumed I shouldn't ask about other topics." Person B said "I would think it's an expert at travel and it just didn't feel right if I asked random things." The title "Travel Book" printed on the cover of the book motivated users to talk only about travel.

Some users realized that the book knew their location, which raised concerns about the personal information stored by the book. Person E said "I would provide the key information that the book would require." Person C said "I don't want it to share my data with others." Users want to choose how their information is treated.

Some users would want to converse with their favorite everyday objects such as a blanket which has been used by them for a long time. Person A said "The blanket can store my memory and sometimes I want to hold and talk with it about my past." The conversation with the travel book is similar to one with a stranger. But there are users' needs to talk with their favorite objects which may hold personal memories.

5 CONCLUSION AND FUTURE WORK

We presented the design fiction narrative and the conversational system called The Talk. Our user test showed a positive acceptance of conversation with everyday objects, and users tried to build a human-like relationship with them. Future work will be conducted to understand longer term relationships with conversational everyday objects. We believe that the results will improve the use of everyday objects.

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